

ALEXANDER C. JENKINS

Kavli Institute for Cosmology and DAMTP | University of Cambridge | Cambridge, UK

acj46@cam.ac.uk | [Website](#) | [Publications](#) | [GitHub](#) | [LinkedIn](#)

ABOUT ME

I'm a theoretical physicist, working at the interface between *cosmology*, *high-energy physics*, and *quantum matter*. My research looks at new ways of probing the fundamental laws of Nature, whether that's using *gravitational waves* as cosmic messengers, or using *quantum technologies* to simulate the early Universe.

EMPLOYMENT

Gavin Boyle Fellow in Cosmology — *University of Cambridge* 2024–present

EPSRC Stephen Hawking Fellow 2025–present

Research fellow hosted in the [Kavli Institute for Cosmology Cambridge](#) (KICC) and the [Department of Applied Mathematics and Theoretical Physics](#) (DAMTP) | Fellow of [Selwyn College](#)

Postdoctoral Research Fellow — *University College London* 2021–2024

Led an international, interdisciplinary project to study vacuum decay with quantum analogue experiments and lattice simulations, as part of the [QSimFP Consortium](#) | Member of the [Cosmoparticle Initiative](#)

EDUCATION

PhD in Theoretical Physics — *King's College London* 2017–2021

Funded by competitive faculty scholarship | [Theoretical Particle Physics and Cosmology Group](#)

'*Cosmology and fundamental physics in the era of GW astronomy*', supervised by [Mairi Sakellariadou](#)

MSci in Astrophysics (Part III) — *University of Cambridge* 2016–2017

[1st class](#) | Ranked 5th in cohort | Elected a Bateman Scholar of [Trinity Hall](#) for 'excellent exam results'

Leonid Grishchuk Internship Programme — *Cardiff University* Summer 2016

Funded summer research internship in the [Gravitational Physics Group](#)

MA in Natural Sciences (Astrophysics) — *University of Cambridge* 2013–2016

[1st class](#) | Elected a Scholar of [Trinity Hall](#)

GRANTS AND FUNDING SECURED (POST-PHD)

- EPSRC Stephen Hawking Fellowship (PI, £506k fEC) 2025
3-year research council fellowship supporting 'visionary scientists working in theoretical physics'
Proposal ranked 2nd out of 26 by EPSRC prioritisation panel
- Gavin Boyle Fellowship, KICC and Selwyn College, Cambridge | 5-year institutional fellowship 2024
- UKRI Quantum Technologies for Fundamental Physics Grant extension (Co-I, £69k) 2023
Wrote successful proposal for 6-month funded extension to my QSimFP research

AFFILIATIONS

Scientific collaborations

* Quantum Simulators for Fundamental Physics (2021–present) * LISA Consortium (2018–present)

* Einstein Telescope (2021–present) * LIGO Scientific Collaboration (2016–2021)

Professional bodies

* Institute of Physics (MInstP; 2018–present) * Royal Astronomical Society (FRAS; 2020–present)

* Junior Member, European Astronomical Society (2020–present)

AWARDS AND ACHIEVEMENTS

- Winner, [Buchalter Cosmology Prize](#) (2nd Prize) | [UCL press release](#) 2023
International award recognising ‘ground-breaking theoretical, observational, or experimental work in cosmology that has the potential to produce a breakthrough advance in our understanding’
- Honorable Mention, [GWIC-Braccini Thesis Prize](#) | nominated for three further thesis prizes 2022
- Best Student Talk Prize at [BritGrav 21](#), sponsored by IoP Publishing 2021
Corresponding paper published in *Classical and Quantum Gravity* as an invited submission
- [King’s Education Award](#) for ‘extraordinary contributions’ to teaching 2020
Also nominated as a ‘Rising Star’ in 2019 (only PhD student nominee in my department)
- Bateman Scholar of [Trinity Hall](#) (Cambridge), recognising ‘excellent exam results’ 2017

RESPONSIBILITIES

Leadership roles in scientific collaborations and networks

- Co-organiser, [CamGW](#) — the Cambridge gravitational-wave network 2025–present
Initiated an inter-departmental network to connect gravitational-wave researchers across Cambridge; organised a successful programme of [meetings](#) with external speakers and 40+ attendees
- Co-organiser, [GW:UK](#) — connecting the UK’s gravitational-wave research community 2025–present
- Group lead, [GUEST](#) (*Gravitational Universe Exploration with Satellite Tracking*) 2025–present
Leading a working group to develop the science case for a proposed F-class ESA mission

Organisation of conferences and seminars

- Lead organiser, *UCL Cosmology/Extragalactic Seminars* 2022–2023
Developed a programme of in-person talks from what had previously been an online-only event due to COVID-19; secured departmental funding for speaker expenses and group lunches
- Co-organiser, [London Cosmology Discussion Meetings \(LCDM\)](#) 2018–2021
Coordinated between five institutions to organise meetings at the Royal Astronomical Society
- Lead organiser, *Theoretical Particle Physics and Cosmology (TPPC) Journal Club* 2021
- Lead organiser, *TPPC Gravity Meetings* 2018–2020
Initiated a regular series of meetings with internal and external speakers on gravitational physics

Refereeing and review work

- Referee for seven research grants, including two ERC Consolidator Grants, 2019–present
one NSF Research Grant, one EPSRC Research Grant, one EPSRC Stephen Hawking Fellowship, one Royal Society Dorothy Hodgkin Fellowship, and one grant for the Polish National Science Centre
- Referee for 32 articles in Physical Review Letters, Nature Astronomy, 2020–present
Astrophysical Journal Letters, Physical Review D, Journal of Cosmology and Astroparticle Physics, European Physical Journal C, The Astronomical Journal, and Universe

Committees and institutional service

- Elector, Selwyn College Mastership Election 2024–2025
Contributed to interviews and selection process for the new head of my Cambridge college
- Selwyn College Governing Body and IT and Data Committee 2024–present
- Student representative, KCL Physics Department Research Committee 2019–2021

TEACHING AND SUPERVISION

University of Cambridge

2024–present

- Delivered three 1hr lectures for 3rd-year *Relativity* (Part II Physics/Astrophysics); student feedback highlighted my ‘very clear and engaging lecturing’
- Supervisor of two MSci students: Teymour Taj and Felix Bowden
- Led small-group ‘supervision’ teaching in *Relativity*, *Statistical Physics*, and *Principles of Quantum Mechanics* for 3rd-year Astrophysicists

University College London

2021–2024

- Lead supervisor of two MSc students: Phoebe Routh ([distinction](#), now a PhD student in Glasgow) and David Moody ([distinction](#) and awarded departmental prize, now working in data science)
- Teaching assistant for 3rd-year *Physical Cosmology*: developed problem sets, delivered problem-solving tutorials, and contributed to marking of final exams

King’s College London

2017–2021

- Winner of a 2020 [King’s Education Award](#), recognising ‘extraordinary contributions’ to teaching
- ‘Rising Star’ nominee in the 2019 King’s Education Awards (only PhD student nominee in Physics)
- Co-wrote lecture notes for 3rd-year *General Relativity and Cosmology*
- Examples class demonstrator for numerous courses, including 4th-year *Astroparticle Cosmology*, 3rd-year *General Relativity and Cosmology*, 2nd-year *Astrophysics*, 1st-year *Mathematics for Physicists*, ...

SELECTED PUBLIC ENGAGEMENT

- Interviewed on ‘[Unexpected Elements](#)’ from the BBC World Service 2025
- Contributed to the art/science exhibition ‘[Cosmic Titans](#)’ at Lakeside Arts, Nottingham 2025
Interacted with artist Conrad Shawcross RA to provide input for his piece ‘*The Blind Proliferation*’; gave an [interview](#) exploring the science behind Conrad’s work, featured in the exhibition and online
- My research and simulations featured in the documentary ‘*Do we live in a multiverse?*’ 2024
Aired on [French](#) and [German](#) TV and more than 5 million combined views on YouTube
- Led maths outreach activities at a local school as part of the [Cambridge NRICH project](#) 2024
- Invited speaker for the [Cambridge Astronomical Association](#) 2024
- YouTube video interview on ‘[Early Universe Cosmology in the Lab](#)’ 2023
- Outreach talk for alumni of UCL’s ‘[Introduction to Astronomy](#)’ course 2023
- Participated in five interviews for media pieces on my paper ‘[Bridging the \$\mu\text{Hz}\$ gap in the gravitational-wave landscape with binary resonance](#)’ 2022
- Maths and physics tutor at Open Tutors London 2017–2020
Co-initiated tutoring programme for University of London students from under-represented groups
- Ran an interactive exhibit on Dark Matter at [Science Gallery London](#) 2019
- Local organiser, [Pint of Science Festival](#) 2018

SOFTWARE AND NUMERICS

- Author of Fortran lattice field theory code `lattice-fvd` and Python code `gw-resonance`
- Experience with advanced numerical methods including, e.g., Fourier and Chebyshev pseudospectral methods and symplectic integration
- Extensive experience in Unix environments (Ubuntu/MacOS), including in HPC settings; extensive Python experience (object-oriented programming; data handling and visualisation; `Jupyter`, `NumPy`, `SciPy`, `h5py`, `Astropy`, `healpy`, `sympy`, ...); other languages and software include Fortran, C++, Mathematica, Git, MATLAB, SageMath, SQL, L^AT_EX (including TikZ), ...

SELECTED PUBLICATIONS

Citation statistics as of 11 March 2026 (data from [INSPIRE-HEP](#)):

Lead-author only	17 papers, 772 citations, h -index = 13
Short author list only	30 papers, 1,545 citations, h -index = 18
All publications	112 papers, 41,677 citations, h -index = 71

Lead-author papers (author list is ordered alphabetically in some cases)

- L1. **ACJ**, H. V. Peiris, and A. Pontzen, *Bubbles in a box: Eliminating edge nucleation in cold-atom simulators of vacuum decay* (2025), *Phys. Rev. A* **112**, 023318, [arXiv:2504.02829 \[cond-mat.quant-gas\]](#)
- L2. **ACJ**, I. G. Moss, T. P. Billam, Z. Hadzibabic, H. V. Peiris, and A. Pontzen, *Generalized cold-atom analogues for vacuum decay* (2024), *Phys. Rev. A* **110**, L031301, [arXiv:2311.02156 \[cond-mat.quant-gas\]](#) | [Letter](#)
- L3. **ACJ**, J. Braden, H. V. Peiris, A. Pontzen, M. C. Johnson, and S. Weinfurtner, *Analog vacuum decay from vacuum initial conditions* (2024), *Phys. Rev. D* **109**, 023506, [arXiv:2307.02549 \[cond-mat.quant-gas\]](#) | [Editor's Suggestion](#)
- L4. A. K.-W. Chung, **ACJ**, J. D. Romano, and M. Sakellariadou, *Targeted search for the kinematic dipole of the gravitational-wave background* (2022), *Phys. Rev. D* **106**, 082005, [arXiv:2208.01330 \[gr-qc\]](#)
- L5. M. R. Mosbech, **ACJ**, S. Bose, C. Boehm, M. Sakellariadou, and Y. Y. Y. Wong, *Gravitational-wave event rates as a new probe for dark matter microphysics* (2023), *Phys. Rev. D* **108**, 043512, [arXiv:2207.14126 \[astro-ph.CO\]](#)
Co-lead author with Markus Mosbech; I developed the core idea and led $\sim 50\%$ of the analysis
[Featured](#) in a [Royal Astronomical Society press release](#) at the 2023 National Astronomy Meeting
- L6. D. Blas and **ACJ**, *Bridging the μHz gap in the gravitational-wave landscape with binary resonance* (2022), *Phys. Rev. Lett.* **128**, 101103, [arXiv:2107.04601 \[astro-ph.CO\]](#)
Awarded a [Buchalter Cosmology Prize](#) (2nd Prize), recognising 'potential for remarkable impact'
Altmetric [attention score](#) of 395, [in the top 0.3%](#) of all publications ever tracked by Altmetric
[Featured](#) in the *Daily Express*, *Physics* magazine, *Big Think*, *SYFY wire*, and 40+ other outlets
- L7. D. Blas and **ACJ**, *Detecting stochastic gravitational waves with binary resonance* (2022), *Phys. Rev. D* **105**, 064021, [arXiv:2107.04063 \[gr-qc\]](#)
- L8. **ACJ** and M. Sakellariadou, *Nonlinear gravitational-wave memory from cusps and kinks on cosmic strings* (2021), *Class. Quant. Grav.* **38**, 165004, [arXiv:2102.12487 \[gr-qc\]](#)
[Invited submission](#) to CQG as winner of the Best Student Talk Prize at [BritGrav 21](#)
- L9. **ACJ** and M. Sakellariadou, *Primordial black holes from cusp collapse on cosmic strings* (2020), [arXiv:2006.16249 \[astro-ph.CO\]](#)

- L10. **ACJ**, J. D. Romano, and M. Sakellariadou, *Estimating the angular power spectrum of the gravitational-wave background in the presence of shot noise* (2019), Phys. Rev. D **100**, 083501, [arXiv:1907.06642 \[astro-ph.CO\]](#)
- L11. **ACJ** and M. Sakellariadou, *Shot noise in the astrophysical gravitational-wave background* (2019), Phys. Rev. D **100**, 063508, [arXiv:1902.07719 \[astro-ph.CO\]](#)
- L12. **ACJ**, R. O’Shaughnessy, M. Sakellariadou, and D. Wysocki, *Anisotropies in the astrophysical gravitational-wave background: The impact of black hole distributions* (2019), Phys. Rev. Lett. **122**, 111101, [arXiv:1810.13435 \[astro-ph.CO\]](#)
- L13. **ACJ**, A. G. A. Pithis, and M. Sakellariadou, *Can we detect quantum gravity with compact binary inspirals?* (2018), Phys. Rev. D **98**, 104032, [arXiv:1809.06275 \[gr-qc\]](#)
- L14. **ACJ**, M. Sakellariadou, T. Regimbau, and E. Slezak, *Anisotropies in the astrophysical gravitational-wave background: Predictions for the detection of compact binaries by LIGO and Virgo* (2018), Phys. Rev. D **98**, 063501, [arXiv:1806.01718 \[astro-ph.CO\]](#)
- L15. **ACJ** and M. Sakellariadou, *Anisotropies in the stochastic gravitational-wave background: Formalism and the cosmic string case* (2018), Phys. Rev. D **98**, 063509, [arXiv:1802.06046 \[astro-ph.CO\]](#)
Featured in PRD’s ‘kaleidoscope’ for Sep 2018

Other selected papers (with summary of my main contributions)

- O1. Y. Zhang, F. Wang, P. Wong, **ACJ**, K. Konstantinou, J. Thywissen, C. Eigen, and Z. Hadzibabic, *Quantum Simulation of Massive Relativistic Fields in 2+1 Dimensions* (2026), [arXiv:2603.08840 \[cond-mat.quant-gas\]](#) | First experimental realisation of 2+1D Sine-Gordon model, enabled by my earlier work demonstrating its viability [L2]; significant contributions to theory, conceptualisation, and experimental design throughout the project
- O2. J. W. Foster, D. Blas, A. Bourgoïn, A. Hees, M. Herrero-Valea, **ACJ**, and X. Xue, *Prospects for gravitational wave and ultra-light dark matter detection with binary resonances beyond the secular approximation* (2025), [arXiv:2504.16988 \[gr-qc\]](#)
 Significant conceptual and methodological contributions throughout project
- O3. J. W. Foster, D. Blas, A. Bourgoïn, A. Hees, M. Herrero-Valea, **ACJ**, and X. Xue, *Discovering μ Hz gravitational waves and ultra-light dark matter with binary resonances* (2025), [arXiv:2504.15334 \[astro-ph.CO\]](#) | Significant conceptual and methodological contributions throughout project
- O4. A. Abac *et al.*, *The Science of the Einstein Telescope* (2025), [arXiv:2503.12263 \[gr-qc\]](#)
 Coordinated and led writing of section on cosmic strings for large collaboration white paper
- O5. L. Zwick, D. Soyuer, D. J. D’Orazio, D. O’Neill, A. Derdzinski, P. Saha, D. Blas, **ACJ**, L. Z. Kelley, *Bridging the micro-Hz gravitational wave gap via Doppler tracking with the Uranus Orbiter and Probe Mission: Massive black hole binaries, early universe signals and ultra-light dark matter* (2024), [arXiv:2406.02306 \[astro-ph.HE\]](#) | Led analysis of stochastic gravitational-wave signals and implications for early-Universe physics, wrote corresponding sections of paper
- O6. N. Kouvatsos, **ACJ**, A. I. Renzini, J. D. Romano, M. Sakellariadou, *Unbiased estimation of gravitational-wave anisotropies from noisy data* (2023), [arXiv:2312.09110 \[astro-ph.CO\]](#)
 Proposed new analysis method that has since been adopted by the LVK, and led theoretical work; informal supervision of PhD student (N. Kouvatsos)
- O7. M. Branchesi *et al.*, *Science with the Einstein Telescope: a comparison of different designs* (2023), JCAP **07**, 068, [arXiv:2303.15923 \[gr-qc\]](#) | Contributed to stochastic background sensitivity analysis for different Einstein Telescope configurations, guiding further development of the proposal
- O8. S. Gasparrotto, R. Vicente, D. Blas, **ACJ**, and E. Barausse, *Can gravitational-wave memory help constrain binary black-hole parameters? A LISA case study* (2023), Phys. Rev. D **107**,

- 124033, [arXiv:2301.13228 \[gr-qc\]](#) | Significant conceptual and methodological contributions; informal supervision of PhD student (S. Gasparrotto)
- O9. P. Auclair *et al.* (LISA Cosmology Working Group), *Cosmology with the Laser Interferometer Space Antenna* (2022), *Living Rev. Rel.* **26**, 5, [arXiv:2204.05434 \[astro-ph.CO\]](#) | LISA white paper; contributed to section on cosmic strings, led analysis of related gravitational-wave anisotropies
- O10. A. I. Renzini, B. Goncharov, **ACJ**, and P. M. Meyers, *Stochastic Gravitational-Wave Backgrounds: Current Detection Efforts and Future Prospects* (2022), *Galaxies* **10**, 34, [arXiv:2202.00178 \[gr-qc\]](#) Invited review article; major contributions to sections on gravitational-wave theory and sources
- O11. N. Bartolo *et al.* (LISA Cosmology Working Group), *Probing Anisotropies of the Stochastic Gravitational Wave Background with LISA* (2022), *JCAP* **11**, 009, [arXiv:2201.08782 \[astro-ph.CO\]](#) LISA review paper; coordinator for ‘topological defects’ section, with further contributions to ‘astrophysical sources’ section
- O12. P. Auclair, J. J. Blanco-Pillado, D. G. Figueroa, **ACJ**, M. Lewicki, M. Sakellariadou, S. Sanidas, L. Sousa, D. A. Steer, J. M. Wachter, and S. Kuroyanagi (LISA Cosmology Working Group), *Probing the gravitational wave background from cosmic strings with LISA* (2019), *JCAP* **04**, 034, [arXiv:1909.00819 \[astro-ph.CO\]](#) | LISA review paper; significant writing contributions throughout
- O13. B. P. Abbott *et al.* (LIGO, Virgo), *Directional limits on persistent gravitational waves using data from Advanced LIGO’s first two observing runs* (2019), *Phys. Rev. D* **100**, 062001, [arXiv:1903.08844 \[gr-qc\]](#) | Led interpretation of observational results in the context of cosmological and astrophysical source models, wrote corresponding section
- O14. B. P. Abbott *et al.* (LIGO, Virgo), *Search for the isotropic stochastic background using data from Advanced LIGO’s second observing run* (2019), *Phys. Rev. D* **100**, 061101, [arXiv:1903.02886 \[gr-qc\]](#) Rapid communication | Led and wrote section on implications for cosmic string models

INVITED TALKS

Total of 50 invited talks across three continents

- | | |
|--|-----------------|
| • Cosmology Seminar
<i>Queen Mary University of London</i> | <i>Apr 2026</i> |
| • Particle Theory and Cosmology Seminar
<i>Swansea University</i> | <i>Mar 2026</i> |
| • Brout-Englert-Lemaître Seminar
<i>Brussels</i> | <i>Feb 2026</i> |
| • Quantum Light and Matter Seminar
<i>Durham University</i> | <i>Nov 2025</i> |
| • Particle Cosmology Seminar
<i>Imperial College London</i> | <i>Oct 2025</i> |
| • Numerical Simulations of Early Universe Sources of Gravitational Waves
<i>Nordita, Stockholm</i> | <i>Aug 2025</i> |
| • Cosmology Seminar
<i>CERN, Geneva</i> | <i>Jul 2025</i> |
| • Gravitational Wave Probes of Physics Beyond Standard Model
<i>University of Warsaw</i> | <i>Jun 2025</i> |
| • Multiverse from Home
<i>Online</i> | <i>Jun 2025</i> |

- **WE-Heraeus Seminar: New Windows on the Universe** *May 2025*
Kitzbühel, Austria
- **QSimFP Community Workshop** *Mar 2025*
University of Nottingham
- **Early Universe from Home** *Feb 2025*
Online
- **Theoretical Particle Physics Seminar** *Feb 2025*
University of Sussex
- **Joint seminar, LMU Cosmology and MPI Quantum Optics** *Jan 2025*
Ludwig Maximilian University of Munich
- **General Relativity Seminar** *Dec 2024*
DAMTP, University of Cambridge
- **Cosmology Journal Club** *Nov 2024*
University of Padova (online)
- **Particle Cosmology Seminar** *Nov 2024*
University of Nottingham
- **Cosmology/Extragalactic Seminar** *Oct 2024*
University College London
- **Cambridge-LMU Meeting** *Oct 2024*
Kavli Institute for Cosmology, University of Cambridge
- **GEMMA2 Workshop** *Sep 2024*
Sapienza University of Rome
- **Majorana-Raychaudhuri Seminar** *Sep 2024*
Kolkata/Salerno (online)
- **Cold atoms and molecules for fundamental physics** *Jul 2024*
Cambridge
- **4th EuCAPT Symposium** *May 2024*
CERN, Geneva
- **British Applied Mathematics Colloquium (BAMC)** *Apr 2024*
Newcastle University
- **Cosmology Lunch Seminar** *Feb 2024*
DAMTP, University of Cambridge
- **Gravitational-Wave Group Meeting** *Jan 2024*
Institute of Cosmology and Gravitation (ICG), University of Portsmouth
- **Oberthaler Group Seminar** *Nov 2023*
Kirchhoff Institute for Physics (KIP), Heidelberg
- **Cosmology from Home** *Jul 2023*
Online
- **Quantum Simulators for Fundamental Physics Workshop** *Jun 2023*
Perimeter Institute for Theoretical Physics, Waterloo, Canada
- **Astrophysics Seminar** *May 2023*
University of Leicester

- **Cosmology Seminar** May 2023
Beecroft Institute, Oxford
- **Theory Group Seminar** May 2023
Astroparticle and Cosmology Laboratory (APC), Paris
- **Cosmology and Relativity Seminar** Apr 2023
Queen Mary University of London
- **London Gravity Meeting** Mar 2023
Royal Society, London
- **‘Dark Matters’ Workshop** Nov 2022
Université Libre de Bruxelles (ULB)
- **London-Oldenburg Relativity Seminar** Nov 2022
University College London/University of Oldenburg (online)
- **ICTP-AP Seminar** Sep 2022
International Centre for Theoretical Physics, Asia-Pacific (online)
- **Quantum Simulators for Fundamental Physics Workshop** Sep 2022
Science Gallery London
- **‘Gravitational-Wave Orchestra’ Workshop** Sep 2022
Université Catholique de Louvain, Belgium
- **Theory Group Seminar** May 2022
Institute of High-Energy Physics (IFAE), Barcelona
- **Quantum Technology Seminar** May 2022
London Centre for Nanotechnology, University College London
- **Cosmology/Extragalactic Seminar** Nov 2021
University College London
- **Theory Group Seminar** Oct 2021
Astroparticle and Cosmology Laboratory (APC), Paris
- **Ibarra Group Seminar** Jul 2021
Technical University of Munich (online)
- **Gravitational Wave Probes of Physics Beyond the Standard Model** Jul 2021
University of Warsaw (online)
- **London Cosmology Discussion Meeting (LCDM)** Dec 2020
Royal Astronomical Society, London (online)
- **Theoretical Cosmology Seminar** May 2020
Institute of Cosmology and Gravitation (ICG), University of Portsmouth (online)
- **Cosmology Seminar** May 2020
Beecroft Institute, University of Oxford (online)
- **London Cosmology Discussion Meeting (LCDM)** Feb 2020
Royal Astronomical Society, London
- **Gravitational Wave Probes of Fundamental Physics** Nov 2019
EuCAPT workshop, Amsterdam
- **Seminar** Feb 2019
Virtual Institute of Astroparticle Physics (online)